LAB REPORT – Bayes Optimal Classifier & ML Text Classification

Title: Text Classification on PubMed RCT Dataset using MNB, Hyperparameter Tuning & Bayes Optimal Classifier
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Course: CSE
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1. Introduction

The purpose of this lab was to implement text classification on the PubMed RCT dataset using:

1. Custom Multinomial Naive Bayes (from scratch)
2. Sklearn Naive Bayes model with hyperparameter tuning
3. Bayes Optimal Classifier (BOC) using soft-voting ensemble

The goal was to compare performance across the three methods in terms of Accuracy, Macro-F1, and confusion matrix.

Tasks performed:

- Data extraction & preprocessing
- Feature engineering with CountVectorizer & TF-IDF
- Implementing Naive Bayes from scratch
- Hyperparameter tuning using GridSearchCV
- Computing posterior model weights using validation set
- Implementing BOC via weighted soft voting
- Evaluating metrics & visualizing confusion matrices

2. Methodology

Custom Naive Bayes Implementation

- Used CountVectorizer with unigrams + bigrams
- Applied Laplace smoothing (alpha = 1)
- Calculated priors and conditional word likelihoods manually
- Used log probabilities to prevent underflow
- Predicted using argmax posterior probability

Sklearn Model + Hyperparameter Tuning

- Implemented TF-IDF + MultinomialNB pipeline
- Tuned hyperparameters:
  - n-grams (1,1) and (1,2)
  - min_df = [1, 2, 5]
  - alpha = [0.1, 0.5, 1.0]
- Used GridSearchCV with F1-macro score

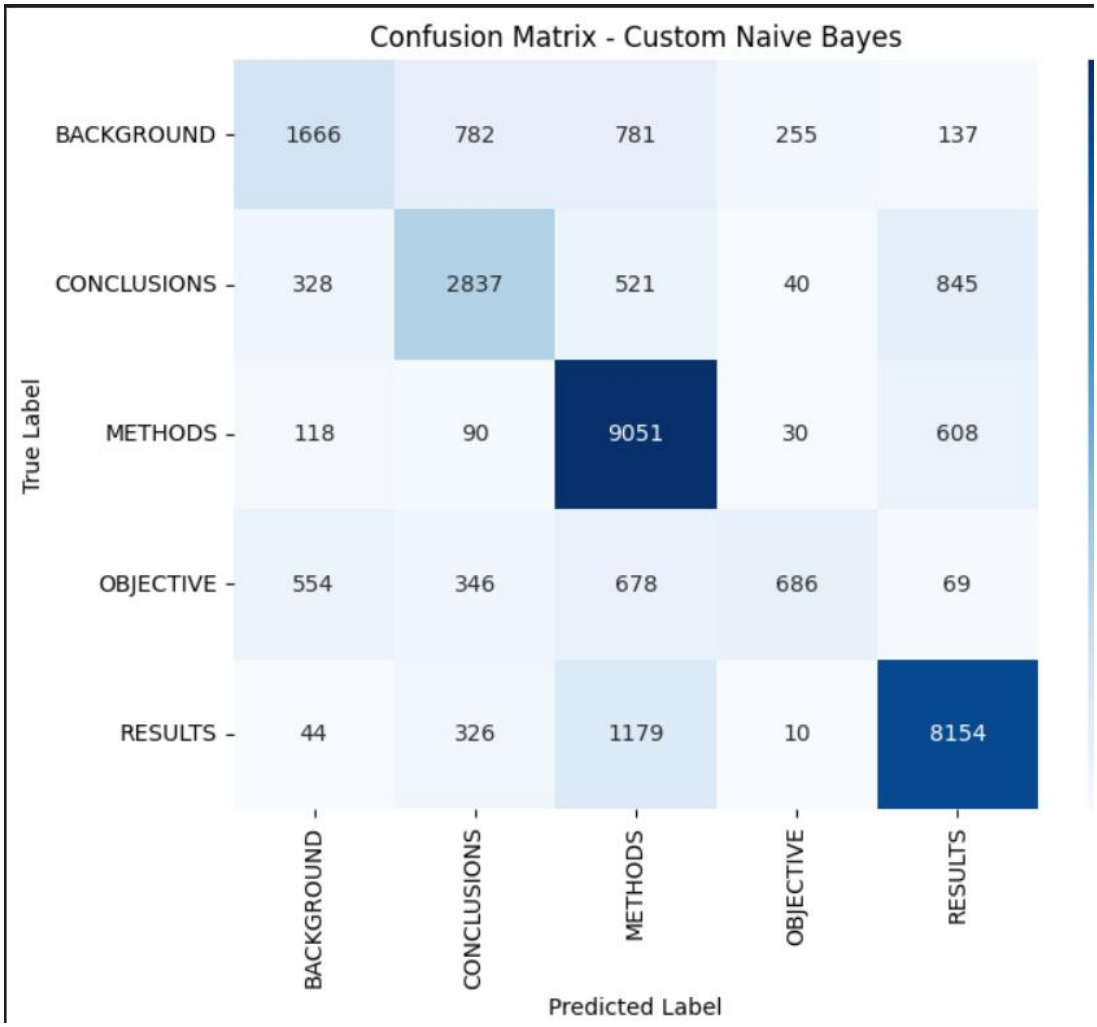
Bayes Optimal Classifier (Approx.)

- Sample size = 10000 + last 3 digits of SRN = 10160
- Five models trained:

- Naive Bayes, Logistic Regression, Random Forest, Decision Tree, KNN
- Computed posterior weights using validation likelihood
- Applied Weighted Soft Voting classifier

3. Results & Analysis

Part A — Custom Naive Bayes Results




```
=== Test Set Evaluation (Custom Count-Based Naive Bayes) ===
Accuracy: 0.7431
              precision    recall  f1-score   support

  BACKGROUND       0.61      0.46      0.53      3621
 CONCLUSIONS       0.65      0.62      0.63      4571
     METHODS       0.74      0.91      0.82      9897
   OBJECTIVE       0.67      0.29      0.41      2333
     RESULTS       0.83      0.84      0.84      9713

    accuracy                           0.74     30135
   macro avg       0.70      0.63      0.64     30135
weighted avg       0.74      0.74      0.73     30135

Macro-averaged F1 score: 0.6446
```

Part B — Sklearn Model + Tuning

Best Hyperparameters:

- ngram_range = (1,2)
- min_df = 5
- alpha = 0.1

```
Training initial Naive Bayes pipeline...
Training complete.

=== Test Set Evaluation (Initial Sklearn Model) ===
Accuracy: 0.6996
              precision    recall  f1-score   support

  BACKGROUND       0.61      0.37      0.46      3621
 CONCLUSIONS       0.61      0.55      0.57      4571
     METHODS       0.68      0.88      0.77      9897
   OBJECTIVE       0.72      0.09      0.16      2333
     RESULTS       0.77      0.85      0.81      9713

    accuracy                           0.70     30135
   macro avg       0.68      0.55      0.56     30135
weighted avg       0.69      0.70      0.67     30135

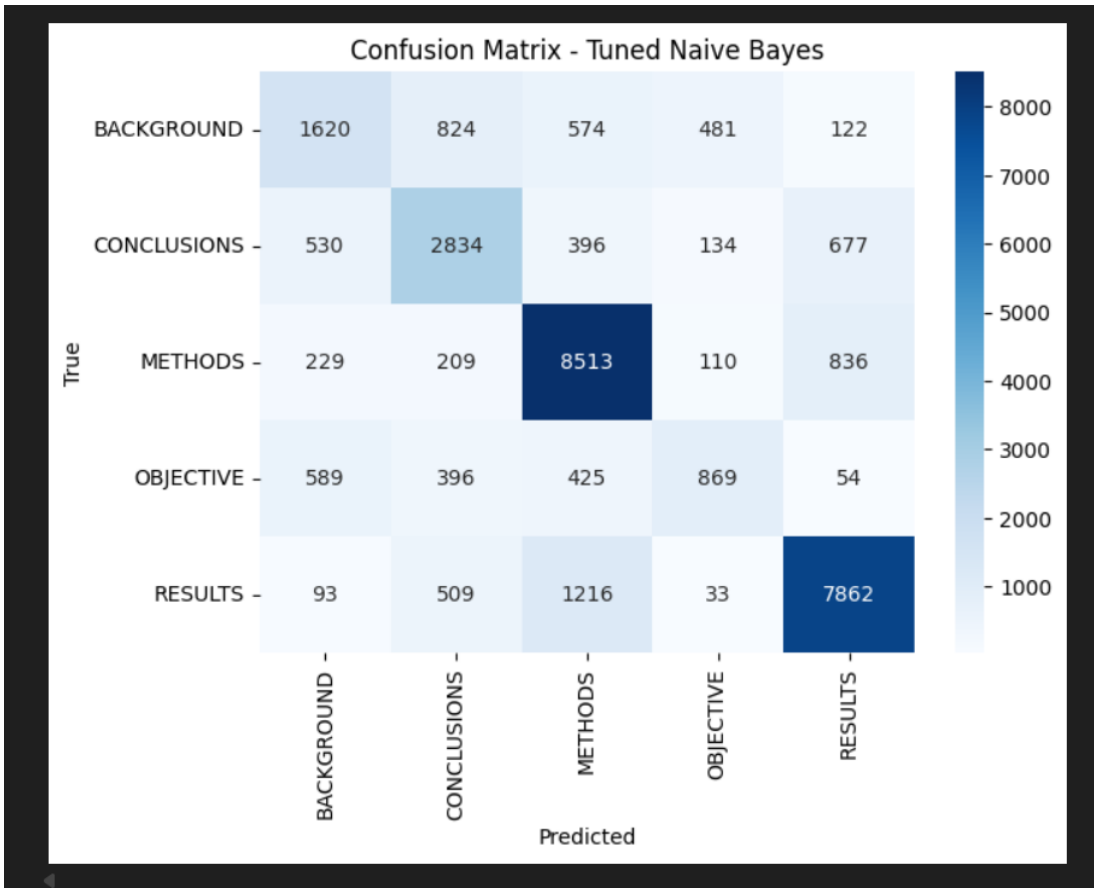
Macro F1: 0.5555

Starting Hyperparameter Tuning on Development Set...
Fitting 2 folds for each of 18 candidates, totalling 36 fits
Grid search complete.

Best Parameters: {'nb__alpha': 0.1, 'tfidf__min_df': 5, 'tfidf__ngram_range': (1, 2
Best Dev F1: 0.6167
...
   macro avg       0.65      0.62      0.63     30135
weighted avg       0.71      0.72      0.71     30135

Macro F1: 0.6313
```

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Part C — Bayes Optimal Classifier
SRN suffix: 160
Dynamic Dataset Size Used: 10160
Posterior Weights:

- Naive Bayes: 0.2322
- Logistic Regression: 0.2456
- Random Forest: 0.2163
- Decision Tree: 0.1655
- KNN: 0.1405

Metric	Score
Accuracy	0.6926
Macro-F1	0.5842

```
Using dynamic sample size: 10160
Actual sampled training size = 10160

Training all base models...
C:\Users\melsh\AppData\Roaming\Python\Python312\site-packages\sklearn\linear_model\_logistic.py:1296: Futu
  warnings.warn(
✅ All base models trained.
C:\Users\melsh\AppData\Roaming\Python\Python312\site-packages\sklearn\linear_model\_logistic.py:1296: Futu
  warnings.warn(

Posterior Weights P(h|D):
NaiveBayes: 0.2322
LogisticRegression: 0.2456
RandomForest: 0.2163
DecisionTree: 0.1655
KNN: 0.1405

Fitting Bayes Optimal Classifier...
C:\Users\melsh\AppData\Roaming\Python\Python312\site-packages\sklearn\linear_model\_logistic.py:1296: Futu
  warnings.warn(
✅ BOC training complete.

=== BOC Final Evaluation ===
Accuracy: 0.6926
Macro F1: 0.5842
              precision    recall  f1-score   support

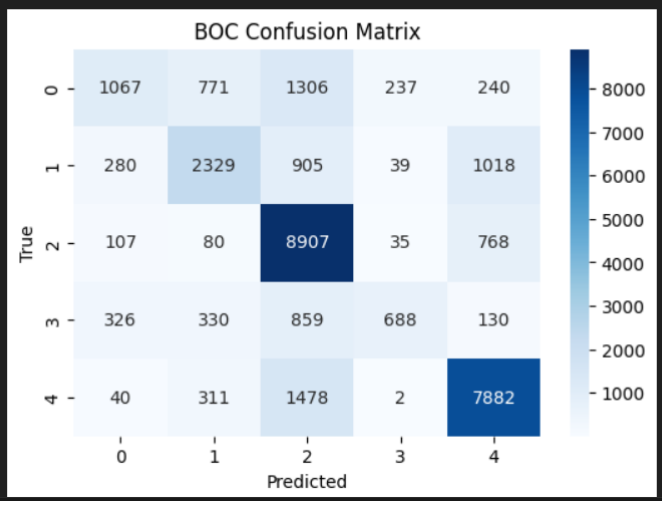
  BACKGROUND       0.59      0.29      0.39      3621
 CONCLUSIONS       0.61      0.51      0.56      4571
     METHODS       0.66      0.90      0.76      9897
   OBJECTIVE       0.69      0.29      0.41      2333
     RESULTS       0.79      0.81      0.80      9713

    accuracy                           0.69     30135
   macro avg       0.67      0.56      0.58     30135
```


```
Macro F1: 0.5842
              precision    recall  f1-score   support

  BACKGROUND       0.59      0.29      0.39      3621
 CONCLUSIONS       0.61      0.51      0.56      4571
     METHODS       0.66      0.90      0.76      9897
   OBJECTIVE       0.69      0.29      0.41      2333
     RESULTS       0.79      0.81      0.80      9713

    accuracy                           0.69     30135
   macro avg       0.67      0.56      0.58     30135
weighted avg       0.69      0.69      0.67     30135
```

4. Discussion

Model	Macro-F1	Remarks
Custom NB	0.6446	Best overall, strong on text data
Tuned Sklearn NB	0.6313	Bigram + tuned alpha improved score
BOC	0.5842	Relatively weaker due to RF/KNN inefficiency on sparse text

Key Observations:

- Naive Bayes excels in bag-of-words text classification
- Logistic Regression also performs well, contributing strongest weight in BOC
- BOC didn't outperform NB since tree-based models struggle with sparse features
- Hyperparameter tuning benefited the NB, but still slightly below custom version

5. Conclusion

- Custom Naive Bayes was the best performing method
- Sklearn tuning improved results but did not surpass manual NB
- Bayes Optimal Classifier provided robust ensemble performance but was hindered by weaker models
- Text classification tasks are best suited for Naive Bayes / Linear models
- Ensembles work best when all base models are individually strong